Structural Selection Across Field Theory and Cosmology: A MAAT Functional Benchmark in Nonlinear and FLRW Toy Models

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Abstract

We extend the MAAT structural selection framework from nonlinear field configurations to simple cosmological histories. The framework separates dynamical admissibility from structural preference through a functional F_{MAAT}, built from equation consistency, balance, controlled activity, connectivity, and robustness [1, 2]. Previous benchmarks in ϕ^4 and Sine-Gordon models showed that F_{MAAT} selects coherent solitonic structures over distorted or chaotic configurations, including at approximately fixed energy. Here we add a flat FLRW scalar-field toy model and show that the same structural logic ranks cosmological histories according to coherent expansion, dynamical productivity, and robustness against pathological regimes. The resulting hierarchy favors slow-roll coherent evolution, then kinetic relaxation, then weakly generative static expansion, while penalizing ghost-like and collapse-like histories. These results provide numerical evidence that structural selection captures information not contained in energy-based or purely dynamical descriptions, and generalizes consistently from static field configurations to cosmological dynamics.

1 Introduction

Standard physics determines which configurations or histories are dynamically allowed. It does not always determine which of those possibilities are structurally preferred or physically realized.

The MAAT structural selection programme proposes a second layer [1, 2]:

$$S_0[\Phi] \Rightarrow \text{dynamical admissibility} \tag{1}$$

$$F_{\mathrm{MAAT}}[\Phi] \Rightarrow \text{structural preference} \tag{2}$$

The selection principle is

$$\Phi_{\text{selected}} = \arg\min_{\Phi} F_{\mathrm{MAAT}}[\Phi], \tag{3}$$

or probabilistically,

$$P[\Phi] \propto \exp[-\beta F_{\mathrm{MAAT}}[\Phi]]. \tag{4}$$

In previous work [1], the framework was tested in nonlinear field systems. In this paper, we extend the same logic to cosmological toy histories.

2 MAAT Structural Functional

The MAAT functional evaluates five structural dimensions [2, 3]:

- H: equation consistency,

- B: balance and constraint consistency,
- S: controlled dynamical activity,
- V: connectivity or temporal coherence,
- R: robustness against pathological behavior.

Each sector is represented as a bounded support score:

$$\Gamma_a = \frac{1}{1+d_a}, \qquad a \in \{H, B, S, V, R\}. \tag{5}$$

The structural cost is

$$F_{\mathrm{MAAT}} = -\sum_a \lambda_a \log(\epsilon + \Gamma_a). \tag{6}$$

In all numerical benchmarks reported in this paper, the sector weights are set uniformly to $\lambda_a = 1$. The associated stability score is

$$\text{Stability} = \min\Big(R, (HBSV)^{1/4}\Big). \tag{7}$$

The stability definition follows a conservative criterion in which robustness acts as a hard constraint, while the geometric mean captures balanced structural support across sectors. The robustness score R acts as a hard cap: no degree of dynamical productivity compensates for a violated robustness condition.

3 Nonlinear Field Benchmarks

3.1 ϕ^4 Model

The scalar field model is

$$S_0[\phi] = \int d^d x \left[\frac{1}{2}(\partial\phi)^2 - \frac{\lambda}{4}(\phi^2 - v^2)^2\right]. \tag{8}$$

The tested configurations include vacua, kinks, domain walls, distorted structures, strongly distorted structures, and chaotic fields. The observed structural ordering is

$$F_{\mathrm{MAAT}}(\text{kink}) < F_{\mathrm{MAAT}}(\text{distorted}) < F_{\mathrm{MAAT}}(\text{strong distorted}) < F_{\mathrm{MAAT}}(\text{chaos}). \tag{9}$$

3.2 Equal-Energy Test

A central test constructs configurations with approximately equal energy. Despite nearly identical energy values, F_{MAAT} distinguishes coherent perturbations from rough noise.

State	Energy	F_{MAAT}
two-bump	0.966605	1.539255
localized bump	0.966578	1.595588
smooth wave	0.966628	1.998289
rough noise	0.966626	12.155561

Table 1: Equal-energy benchmark. All configurations have approximately the same energy, but strongly different structural scores. This demonstrates that structural selection is not equivalent to energy minimization.

3.3 Sine-Gordon Model

The Sine-Gordon model is defined by $-\phi'' + \sin(\phi) = 0$, with soliton solution $\phi(x) = 4\arctan(e^x)$ [4]. The MAAT ranking is

$$F_{\mathrm{MAAT}}(\text{soliton}) < F_{\mathrm{MAAT}}(\text{distorted}) < F_{\mathrm{MAAT}}(\text{strong distorted}) < F_{\mathrm{MAAT}}(\text{chaos}). \tag{10}$$

State	Energy	F_{MAAT}
soliton	7.998469	0.886317
distorted soliton	8.583749	1.509674
strong distorted soliton	18.361648	6.654255
chaos	7883.320242	32.013829

Table 2: Sine-Gordon structural benchmark. The soliton is structurally preferred.

4 Cosmological Toy Model

We test the framework on flat FLRW scalar-field histories [7, 9]. The equations are

$$H^2 = \frac{\rho}{3}, \qquad \rho = \frac{1}{2}\dot{\phi}^2 + V(\phi), \qquad \ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0. \tag{11}$$

The main plateau potential is

$$V(\phi) = V_0\left[1 - \tanh^2\left(\frac{\phi}{\Delta}\right)\right] + \epsilon. \tag{12}$$

To test pathological behavior, we include a negative potential toy case $V_{\mathrm{neg}}(\phi) = 0.2 - 0.08\phi^2$ and a ghost-like kinetic sign reversal.

5 Cosmological Structural Diagnostics

For cosmological histories the MAAT components are:

- H: Friedmann consistency,
- B: positive density and finite equation of state,
- S: productive expansion and controlled activity,
- V: temporal smoothness and coherence,
- R: robustness against ghost, collapse, kinetic domination, or invalid density.

The tested histories are: slow-roll coherent expansion, kinetic relaxation, weakly generative static expansion, collapse from negative potential, and ghost-like accelerated expansion.

6 Cosmology Results

The observed ranking is

$$\text{slow roll} < \text{kinetic relaxing} < \text{static dead} < \text{ghost like} < \text{collapse}. \tag{13}$$

This ordering follows directly from the combined effect of equation consistency, controlled activity, and robustness penalties. In particular, ghost-like solutions are penalized due to kinetic instability despite producing many e-folds, while collapse histories fail density positivity and Friedmann consistency simultaneously.

History	F_{MAAT}	Stability	e-folds
slow-roll coherent	0.226081	0.945047	20.826305
kinetic relaxing	0.298763	0.929287	13.147509
static dead	0.578187	0.865414	3.522249
ghost-like	2.781332	0.090909	34.408448
collapse negative potential	40.485202	0.000000	0.006106

Table 3: Cosmological MAAT ranking. Slow-roll coherent expansion is preferred. The ghost-like case produces 34 e-folds but is penalized by $R = 0.09$, illustrating that dynamical productivity without robustness is not structurally selected.

7 Figures

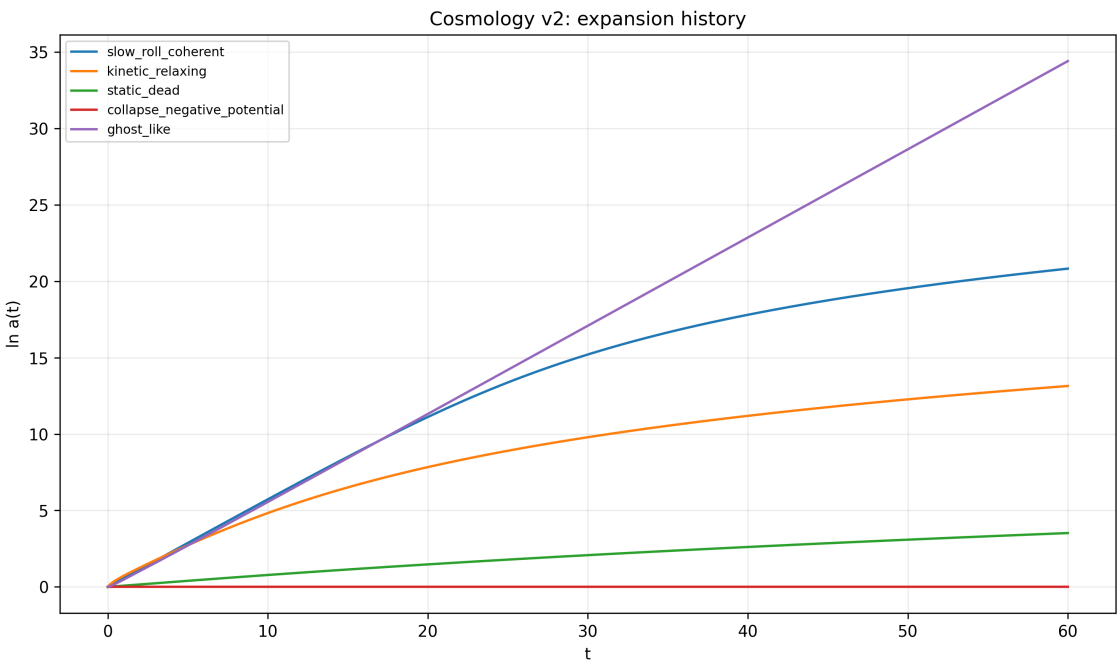

Figure 1: Expansion histories. Ghost-like expansion proceeds rapidly but is penalized by low robustness.

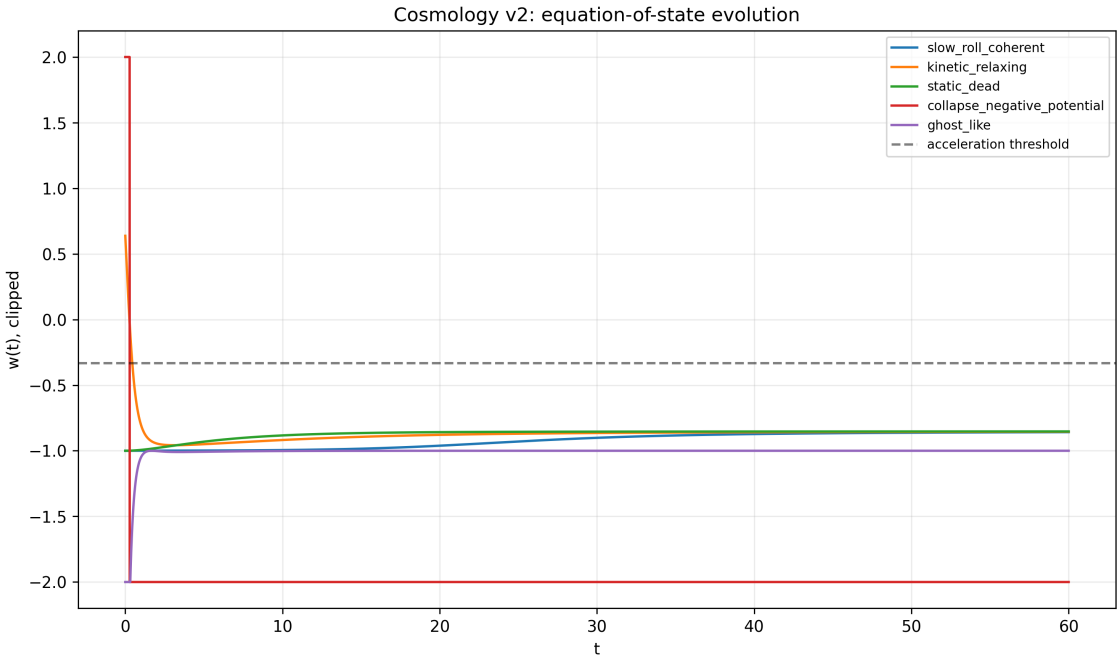

Figure 2: Equation-of-state evolution. Collapse and ghost-like regimes show pathological behavior relative to coherent slow-roll evolution.

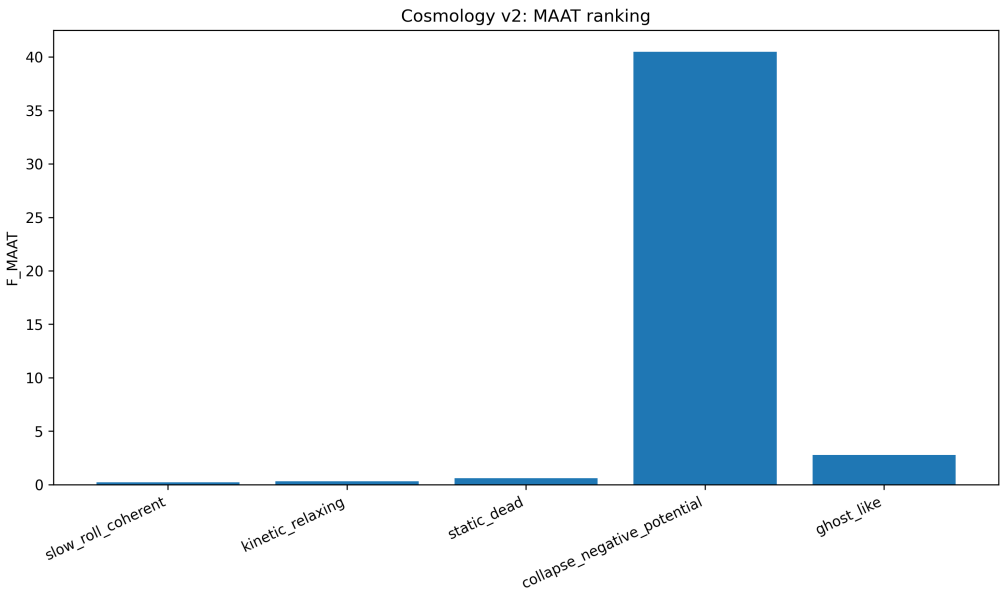

Figure 3: MAAT ranking of cosmological histories. Lower F_{MAAT} indicates stronger structural preference.

8 Discussion

The field-theory tests show that F_{MAAT} identifies coherent structures across different nonlinear systems. The equal-energy test shows that the functional is not reducible to ordinary energy minimization.

The cosmological extension illustrates the role of the stability rule (7): even if a history is dynamically productive, it is rejected when robustness fails. The ghost-like case produces

34 e-folds but achieves $R = 0.09$ (see stability rule (7)), yielding $F_{\mathrm{MAAT}} = 2.78$, far above the preferred slow-roll case at 0.23.

The cosmological results can be interpreted as a dynamical extension of the structural selection principle observed in static field configurations: coherent attractor-like dynamics replaces solitonic coherence as the structurally preferred regime.

If this behaviour persists in more realistic models, structural selection could act as an additional organizing principle for physically realized cosmological histories.

The consistency of the structural ordering across independent model classes — ϕ^4, Sine-Gordon, and FLRW cosmology — suggests that the observed behaviour is not model-specific but reflects a more general structural property of dynamical systems.

In field theory, the functional measures spatial coherence of configurations, while in cosmology it evaluates temporal coherence of dynamical histories. The same architecture applies in both cases without modification.

9 Code and Reproducibility

The numerical pipeline used for the cosmological toy benchmark is available in the public repository under the `experiments/` directory:

github.com/Chris4081/structural-selection-principle/tree/main/experiments.

The cosmology-specific reproducibility bundle is:

`experiments/cosmology_structural_selection/`.

It contains the executable script

`maat_cosmology_toy_v2.py`,

which generates:

`maat_cosmology_toy_v2_results.json`

and the figure directory:

`maat_cosmology_toy_v2_plots/`.

The default run can be reproduced with:

```
python3 maat_cosmology_toy_v2.py
```

The script uses fixed parameters specified in the source file and writes the ranking, all diagnostic sector scores, e-fold counts, and generated plot names to the JSON output. The intended reproducibility loop is:

$$\text{FLRW toy histories} \to \text{structural diagnostics} \to \text{JSON results} \to \text{plots}.$$

10 Limitations

This work remains exploratory. The models are simplified, the diagnostics are operational proxies, and the framework is not yet derived from a microscopic theory [2]. The results do not establish a complete theory of nature.

The present claim is narrower:

> A single structural selection architecture can consistently rank coherent, distorted, chaotic, and pathological configurations across multiple toy systems.

11 Conclusion

We provide numerical evidence that MAAT structural selection generalizes from nonlinear field configurations to simple cosmological histories. The main findings are:

- coherent structures are selected over chaotic configurations,
- structural selection differs from energy minimization,
- at fixed energy, MAAT distinguishes structural quality,
- the same logic ranks cosmological histories by coherent expansion and robustness.

These results support further investigation of structural selection as an organizing principle in field theory and cosmology.

References

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